

Evaluating PFOS and PFOA Standards: Health Risks, Population Exposure, and Regulatory Implications

Per- and polyfluoroalkyl substances (PFAS), including perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA), are persistent environmental pollutants with significant health implications. This study uses a novel Relative Health Indicator (RHI) metric to assess potential health risk reductions under different regulatory standards by normalizing contaminant risk to a single scale. This metric enables us to quantify a “meaningful opportunity” for a potential regulation and compare across previously determined regulatory contaminant values to help establish a minimum threshold for identifying significant regulatory opportunities. Using this metric, we are also able to compare the U.S. Environmental Protection Agency's (USEPA) 2024 enforceable maximum contaminant levels (MCLs) with thresholds from governmental and state agencies. When the USEPA proposed a health advisory level of 70 ng/L in 2016, the population considered exposed under this metric was relatively low, 1.44% for PFOS and 0.18% for PFOA. However, with the updated MCL set at significantly lower values of 0.02 ng/L for PFOS and 0.004 ng/L for PFOA, the number of people classified as exposed has substantially increased to 19.91% for PFOS and 17.49% for PFOA. States like California, Connecticut, and Illinois, adhering to USEPA’s strict PFAS thresholds, report higher exposures, while states with higher thresholds report fewer. One key challenge is that detection methods often have minimum reporting limits that exceed low MCLs, making it difficult to confirm concentration and compliance. This risk assessment highlights the trade-off between public health protection and regulatory burden, emphasizing the need for better detection technologies and balanced policies for public safety.

Keyword: Health Risk, MCL, PFOA, PFOS, RHI metric, Risk Assessment

Method 1: Exclude NAs – Raw data for PFOS/PFOA: 143629 obs. = 5712 report + 143629 NAs.
Data skewed right, mean = 0.0194 ug/L.

```
— Data Summary —
Name          values
Number of rows occurrence_pfos_pfoa$Ana1... 149341
Number of columns 1

Column type frequency:
  numeric          1

Group variables   None

— Variable type: numeric —
skim_variable  n_missing complete_rate  mean    sd    p0    p25    p50    p75  p100 hist
1 data          143629      0.0382 0.0194 0.125 0.004 0.005 0.007 0.0125 7  █
```

Table 1. Current (2024) USEPA approved methods to detect PFOS and PFOA used in UCMR3/UCMR4.

MethodID	MRL	Units	MRL	Units	MRL	Units
EPA 533	4	ng/L or ppt	0.004	µg/L or ppb	0.000004	mg/L or ppm
EPA 537	20	ng/L or ppt	0.02	µg/L or ppb	0.00002	mg/L or ppm
EPA 537	40	ng/L or ppt	0.04	µg/L or ppb	0.00004	mg/L or ppm

(a)

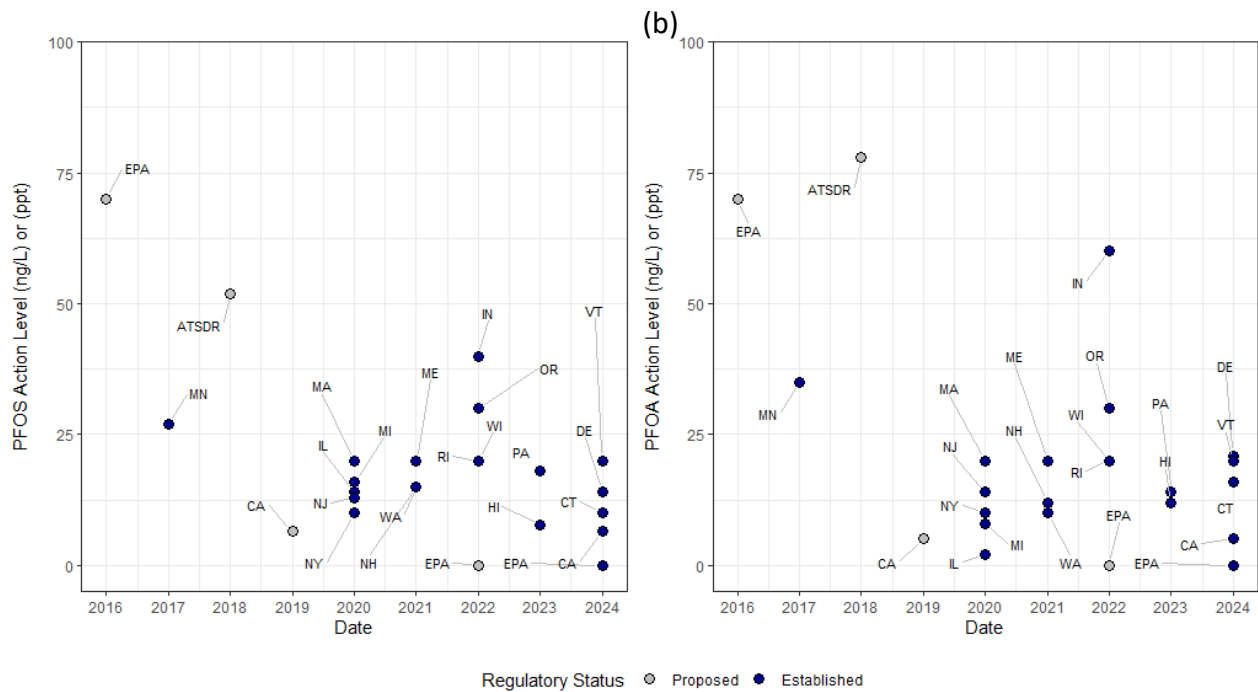


Figure 1. Proposed and established regulatory thresholds for (a) PFOA and (b) PFOS by governmental and state agencies. In 2016 USEPA proposed 70 ng/L as health advisory levels for PFOS and PFOA. In 2024, USEPA published the final enforceable MCL, 0.02 ng/L for PFOS and 0.004 ng/L for PFOA.

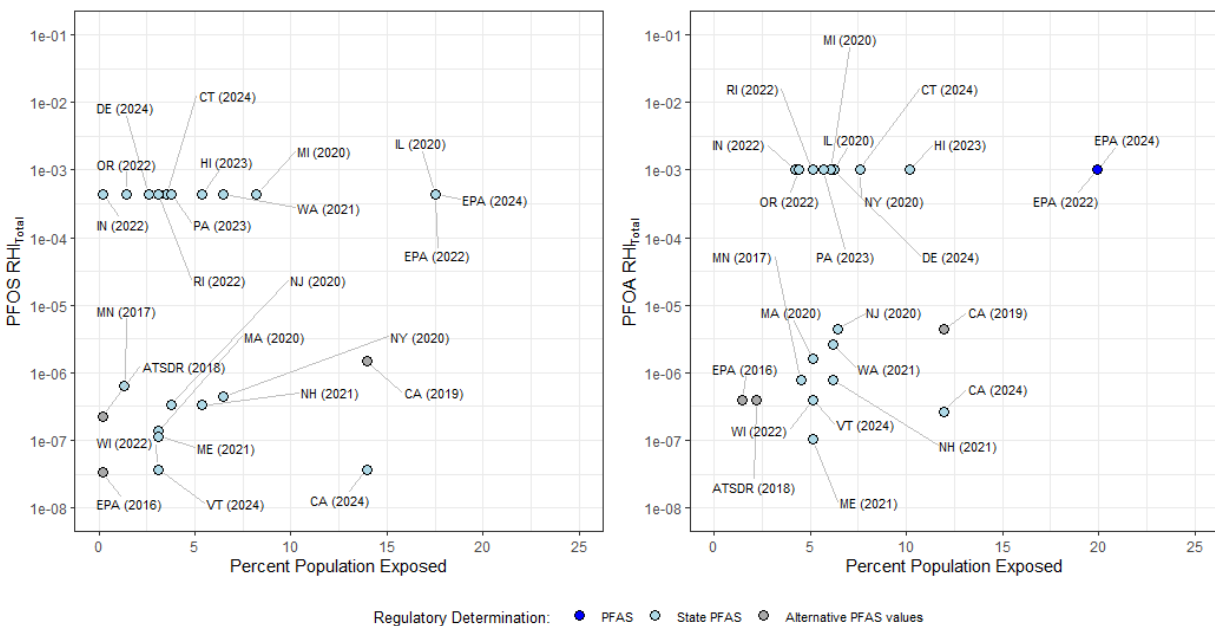


Figure 2. Total RHI and percent population exposed to threshold concentrations proposed (excluding NAs).

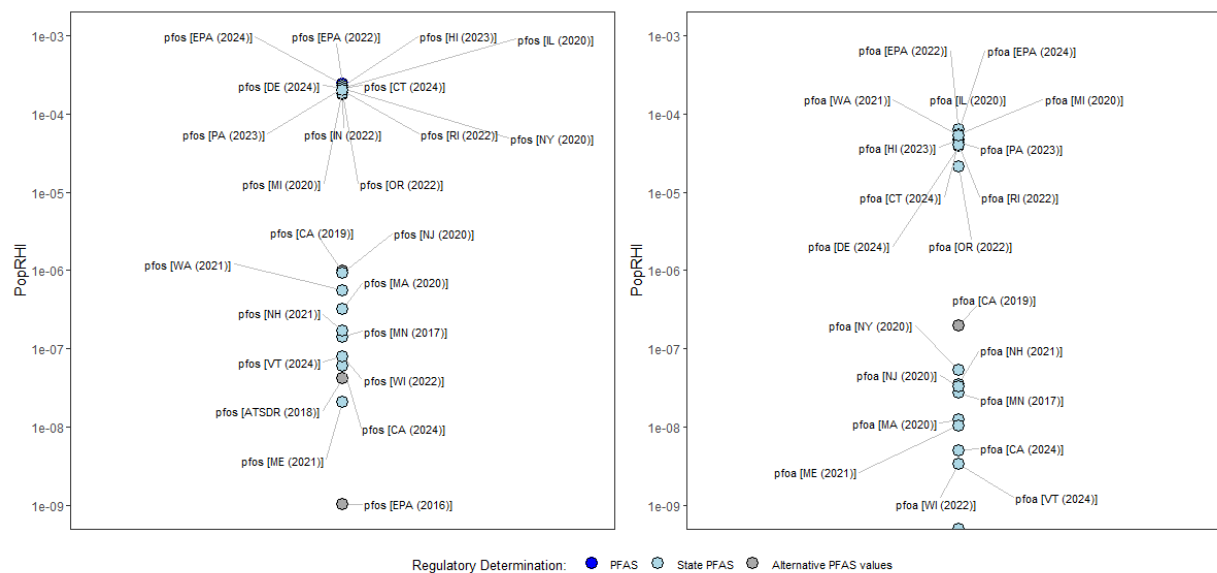


Figure 3. Population RHI for each regulatory determination (excluding NAs).

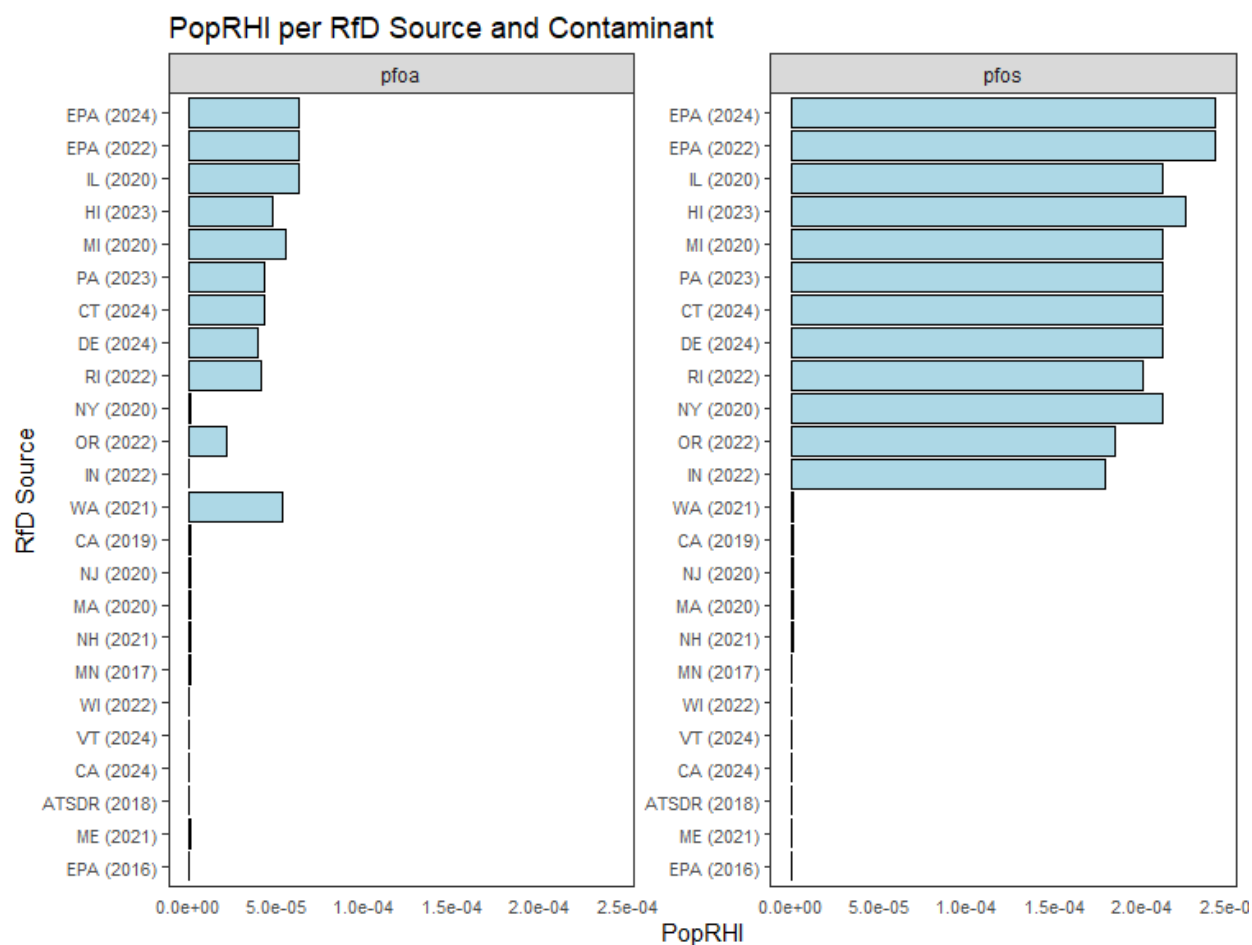


Figure 4. Population exposure by RfD Source for each PFOS and PFOA regulation (excluding NAs).